

Real-time Manufacturing Vision Inspection

ADR | 8/27/2026 | Status: DRAFT - Architect review required

DECISION

Traditional Service Architecture

Azure | Score 97/100 | Confidence 79% (Medium)

\$3,901 expected monthly cost

1. Context

A manufacturing company needs to inspect 100,000 products per day using cameras on production lines. The system must detect scratches, cracks, missing components, and incorrect assembly. Inference must complete within 200 ms, continue operating when internet connectivity is unavailable, and run close to the factory floor. The solution must support GPU-based edge deployment and integrate with existing industrial equipment

2. Architecture-Significant Requirements

SCALE	Enterprise (50K+ users)
AI REQUIREMENT	Let AI Determine
LATENCY OBJECTIVE	P95 within 3,000 ms
DEMAND	Under 100K monthly; 3 requests/second peak
DATA SENSITIVITY	High / Confidential
COMPLIANCE	None specified
SECURITY CONTROLS	Standard controls
INTEGRATIONS	None specified

3. Decision

Adopt Traditional Service Architecture (Rules-based transactional processing) on Azure. The selected option has a deterministic weighted score of 97/100 and is eligible under all mandatory capability gates.

- Lowest operational complexity
- Predictable behavior and cost

4. Solution Components

Component		
Load Balancer		
Identity & Access		
Application Services		

Rules Engine		
Message Queue		
Relational DB		
Cache		
Monitoring		

5. Scoring and Evidence

Dimension				
Architecture Fit				
Performance				
Security				
Scalability				
Complexity				
Ai Fit				
Maintainability				
Operational Fit				

6. Cost Estimate

LOW ESTIMATE	\$3,121 / month
EXPECTED ESTIMATE	\$3,901 / month
HIGH ESTIMATE	\$5,071 / month
PRICING BASIS	Deterministic catalog estimate; validate against regional provider pricing and measured usage before approval
Cost area	
Compute	
Data	
AI	
Integration	
Security	
Monitoring	

7. Alternatives Considered

Architecture				
Event-Driven ML				
Hybrid ML + Rules				
Workflow + AI				
Retrieval-Augmented Generation				
Agentic Orchestration				

8. Risks and Mitigations

Severity		
Medium		
Low		
Low		

9. Trade-offs and Consequences

- Performance: 57ms higher estimated P95 latency. Alternative: Event-Driven ML.
- Security: 1 points lower security score. Alternative: Hybrid ML + Rules.

Consequences: the team accepts the operational characteristics, cost range, and delivery complexity of Traditional Service Architecture. Recalculate this ADR when requirements, catalog evidence, provider pricing, or scoring weights change.

10. Assumptions and Approval

- Camera images are available at inspection points
- Existing industrial equipment exposes integration interfaces
- Edge hardware can host GPU-based workloads
- Inference latency budget applies per product image
- Local storage/networking supports offline continuity
- Inspection results are used for pass/fail or routing decisions

Architecture owner / Date

Business owner / Date